

Databases and Information Systems DB Assignment Report

Description

This database allows a national network of hospitals to track possible candidates applying for positions with the healthcare system. It includes tracking their personal details and skills, while also managing interview times, dates and whether the individual was accepted or rejected. The database implores the use of many-to-many relationships between positions/candidates and skills. This allows multiple positions/candidates to be applied to one skill and for one skill to be applied to multiple positions/candidates.

Database includes:

- Hospital Details – included information about each hospital in the network (i.e., Hospital ID, name, address, phone number)
- Candidate Details - included information about each candidate (i.e., Candidate ID, name, address, phone number)
- Skills – contained different skills which could be selected by candidates (skill ID, skill name)
- Candidate skills – Linked the Candidate ID to Skill ID.
- Position Details – Provided details about the position that is being applied for (i.e., what type of job it was, where it was, and skills required).
- Position skills – Used to link skills required for specific roles.
- Interview Details – Stored the candidates interview details including the time, date and whether they were offered or not.

Assumptions and design

Many assumptions and design choices were taken into consideration during the development of the database.

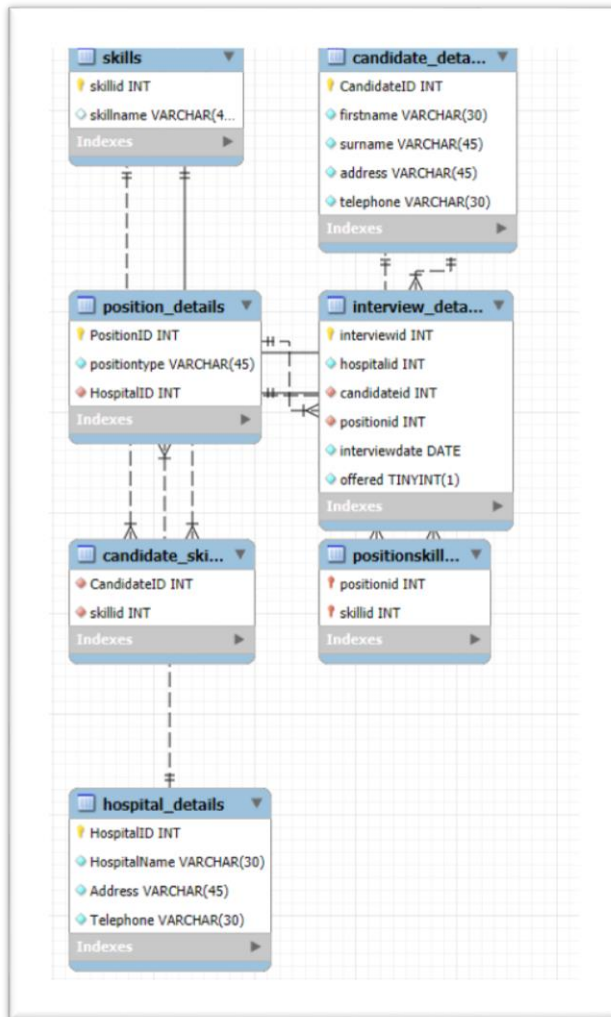
- Skills were stored in a separate table to allow candidates to select multiple skills that apply to them and to ensure that many skills could be applied to many candidates and vice versa.
- Interviews were linked directly to hospitals, candidates and positions as multiple hospitals would be in the network and could possibly have the same details.
- When designing the interview details, I used a Boolean attribute to just display true or false. If the application is pending the box can just be left as null.
- The position skills table was used as a many-to-many relationship between positions and skills
- The candidate's skills table was used as a many-to-many relationship between candidates and skills.

For all of the foreign key relationships I used

- On delete – No action
- On Update – No action

I chose this as in a real-world system, there could be human error which could result in the loss of important data about a candidate or indeed a hospital. Also allows room for updates to the system at a later date. For these reasons I chose to avoid cascade actions for data safety reasons.

ER Diagram



I used Windows 11